Precipitation and Net Ionic Equations

Solubility rules, spectator ions and gravimetric analysis

Name _____ Class _____ Date _____

SOLUBILITY RULES AND DATA

Soluble: Group 1, NH_4^+, nitrates · chlorides (not Ag^+, Pb^{2+}) · sulfates (not Ba^{2+}, Pb^{2+}; Ca^{2+} slightly)

Insoluble: carbonates, phosphates, hydroxides (except Group 1, NH_4^+; $Ba(OH)_2$ soluble)

Molar masses (g/mol): AgCl 143.32 · $BaSO_4$ 233.39 · PbI_2 461.0 · NaCl 58.44 · Cl 35.45

Include state symbols. Check atoms and charge in every net ionic equation. Show units in every step.

A. Solubility

1. Soluble or insoluble? (a) $CaCO_3$ (b) $(NH_4)_2SO_4$ (c) $PbCl_2$ (d) $Cu(NO_3)_2$ (e) $Fe(OH)_3$ (f) K_3PO_4

B. Predicting precipitates

2. Predict whether a precipitate forms when solutions of these are mixed. Name any precipitate. (a) $CaCl_2 + Na_2CO_3$ (b) $KOH + NaNO_3$ (c) $CuSO_4 + NaOH$ (d) $AgNO_3 + KI$

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C. Writing equations

3. For the reaction of barium chloride with sodium sulfate, write (a) the balanced molecular equation (b) the complete ionic equation (c) the net ionic equation. (d) Identify the spectator ions.

4. Write the net ionic equation for each reaction that occurs in question 2.

5. Write the net ionic equation for the neutralization of nitric acid by potassium hydroxide, and explain why it is the same as for hydrochloric acid and sodium hydroxide.

D. Calculations

6. Excess barium chloride is added to 50.0 mL of a sodium sulfate solution. The dried precipitate of $BaSO_4$ has a mass of 0.466 g. Calculate the concentration of sulfate in mol/L.

7. Excess lead(II) nitrate is added to 20.0 mL of 0.120 mol/L potassium iodide. Calculate the mass of lead(II) iodide that forms.

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E. Challenge

8. A 0.300 g sample of impure sodium chloride is dissolved and treated with excess silver nitrate, giving 0.717 g of $AgCl$. Calculate (a) the mass of chlorine in the sample (b) the percent chlorine by mass (c) the percent purity of the NaCl, assuming the impurity contains no chlorine.

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